



Formaldehyde-free self-polymerization of lignin-derived monomers for synthesis of renewable phenolic resin

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ABSTRACT

Most phenolic resins are synthesized with non-renewable petroleum-based phenol and formaldehyde, which have adverse effects on the environment and human health. To achieve green and sustainable production of phenolic resins, it is important to replace non-renewable toxic phenol and formaldehyde. Herein, a new strategy was proposed to completely replace phenol and formaldehyde, using lignin-derived monomers to synthesize renewable phenolic resins. Lithium aluminum hydride was utilized to reduce lignin-derived monomers, including vanillin, methyl vanillate, and syringaldehyde, to generate the corresponding vanillyl and syringic alcohol. With oxalic acid as the catalyst, vanillyl and syringic alcohol could be polymerized to phenolic resins without using formaldehyde. The structure of the phenolic resins based on lignin-derived monomers was analyzed by Fourier transform infrared spectroscopy and ¹³C and ³¹P nuclear magnetic resonance spectroscopy. Differential scanning calorimetry and thermogravimetric analysis were performed to characterize the thermal properties of the phenolic resins. The phenolic resins based on lignin-derived monomers exhibited excellent adhesion strength (6.14 MPa), glass transition temperature (*T*_g) (107–115 °C), and thermal stability, and its performance was similar to that of the commercial Novolak phenolic resin. This study presents a promising green and sustainable approach to synthesize renewable phenolic resins based on lignin-derived monomers without using formaldehyde.

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1. Introduction

Phenolic resins are widely used in a broad range of applications such as electronics, adhesives, carbon foams, molding compounds, fire-resistant and fiber-reinforced composites among others, owing to their low cost, thermostability, and chemical durability [1,2]. Traditionally, non-renewable petroleum-based phenol and formaldehyde are used for the phenolic resin synthesis, which have adverse effects on the environment and human health. Replacing phenol and formaldehyde with sustainable and environmentally friendly materials has attracted much interest in recent decades [3].

Lignin, as the least utilized biomass fraction, has gained increasing attention for its use as raw material in the production of bio-based materials and chemical products, because of its green, non-toxic, and renewable properties [4–7]. Lignin is composed of aromatic monomers (sinapyl alcohols (S), coniferyl alcohol (G), and p-coumaryl alcohols (H)), thus providing an alternative to petroleum-based phenol in phenol formaldehyde (PF) resin synthesis. For example, Kraft, alkali, and organosolv lignin rich in hydroxyl, aldehyde, and carboxylic acid groups have been extensively used as precursors for bio-phenolic resins

synthesis [8,9]. However, the replacement ratio of phenol is limited due to fewer reactive sites and steric hindrance in lignin [10]. In addition, the polymer structures vary depending on the plant type, extraction process and chemical linkage, which makes it difficult to obtain consistent properties [11]. To overcome these limitations, lignin-derived guaiacol and creosote were utilized to synthesize phenolic resins, which enabled the molecular design, structural modification and improved performance of the resulting polymers [12,13]. However, for these phenolic resins synthesis, toxic formaldehyde is essential.

With depleting fossil feedstock and increasingly restrictive health and safety regulations, the substitution of formaldehyde becomes a major concern for the phenolic resin industry. Furanyl compounds (5-hydroxymethyl furfural and furfural) can be derived from cellulose and hemicellulose [14], serving as common building blocks in the synthesis of polymers [15–17]. Hence, replacing the commonly used formaldehyde with furanic compounds is a promising approach to provide formaldehyde-free materials [18,19]. Xu et al. reported a one-pot synthesis for formaldehyde-free phenolic resins, using glucose-derived HMF as a sustainable formaldehyde substitute [20]. For these researches, highly active phenol was used as raw materials commonly. Compared with phenol, the lignin-derived monomers are difficult to couple to these slightly active aldehydes (5-hydroxymethyl furfural and furfural), due to the low reactivity and steric hindrance [21].

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